Raja Babu

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SKILLS

- **Language:** Python, C++, Java, SQL
- **Web & Backend:** HTML, Tailwind CSS, Flask, Spring Boot, Streamlit
- **AI & Data:** Machine Learning, Large Language Models (LLMs), Data Pipelines
- **Core Concepts:** Data Structures, Algorithms, System Design, Computer Networks
- **Tools & Platforms:** Git & GitHub, Docker, MongoDB, Automation Anywhere
- **Soft Skills:** Problem-Solving, Leadership, Adaptability

PROJECTS

Facial Attendance System with Liveness Detection | GitHub Mar 2026

- Designed and developed an AI-powered facial attendance system using Spring Boot and MongoDB Atlas with vector search.
- Implemented 512-dimensional face embedding generation using FaceNet and integrated Python AI services with Java backend via REST APIs.
- Built cosine similarity-based face matching pipeline for real-time identity verification.
- Integrated ONNX-based anti-spoofing (liveness detection) model to prevent photo and screen replay attacks.

Tech Stack: Java, Spring Boot, MongoDB Atlas (Vector Search), Python, FaceNet, InsightFace, ONNX Runtime, OpenCV, REST API, Cosine Similarity

End-to-End MLOps Network Security System with ETL Pipelines | GitHub Dec 2025

- Designing and implementing an end-to-end ETL pipeline to ingest, clean, validate, and transform raw network traffic data for security analysis.
- Building a modular training and evaluation pipeline with reproducibility and experiment tracking.
- Containerizing the ML pipeline using Docker and integrating CI/CD workflows for automated testing and deployment.
- Implementing model monitoring and data drift detection as part of the MLOps lifecycle.

Tech Stack: Python, Pandas, NumPy, Scikit-learn, XGBoost, MLflow, PySpark, Mongodb atlas, Docker, FastAPI

Real-Time Heart Rate Estimation | GitHub July 2025

- Developed a real-time heart rate estimation system using machine learning techniques to process physiological data efficiently.
- Designed and optimized models for accurate heart rate detection from sensor data streams.
- Led the development of a real-time heart rate estimation system using deep learning, enabling non-contact vital sign monitoring through facial video analysis with up to 5.39 MPE.
- Collaborated on integrating the ML pipeline with user-facing interfaces for seamless live monitoring.

Tech Stack: Python, NumPy, Pandas, Scikit-learn, TensorFlow, OpenCV, Flask, DeepPhys

CERTIFICATES

- Certified Essentials Automation Professional by Automation Anywhere — Nov 2025
- Java Programming by Neocolab — May 2025
- CPP Programming by Neocolab — Dec 2024
- Data Structure and Algorithms by Neocolab — Dec 2024

EXTRA-CURRICULAR ACHIEVEMENTS

- Led 15+ Students in Python training, boosting problem solving skills among participants, resulting in more proficient in coding.
- Boosted engagement by designing accessible and practical learning resources in Python and SQL for self-guided learners.

EDUCATION

Lovely Professional University Punjab, India
Bachelor of Technology - Computer Science and Engineering: **CGPA: 7.13** Since August 2023

Govt Co-ed Sr Sec School New Delhi, India
Intermediate: **Percentage: 65** April 2021 - March 2023

Matriculation: **Percentage: 75** April 2020 - March 2021